

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch : Mechanical Engg. Semester :IV Subject Code & Name: BTMEC404_Y19 Numerical Methods in Mechanical Engineering Max Marks: 60 Date:23/01/2024 Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	Write the rules of significant figure. Find the no of significant digits in following : 345.76, 0.035, 2.100×10^3 , π (pi)	CO1	6	
B)	Repeated observations in an experiment gave the values 1.29, 1.33, 1.34, 1.35, 1.32, 1.36, 1.30, and 1.33. Calculate the mean value, absolute error, relative error, and percentage error.	CO1	6	
C)	Define error propagation. Explain analytical approach for addition and multiplication of two quantities using error propagation.	CO1	6	
Q.2	Solve Any Two of the following.		12	
A)	Find the positive root between 2 and 4 which satisfies $x^3 - 9x + 1 = 0$ to three decimal places by Bisection method.	CO2	6	
B)	Find the positive root of $f(x) = 2x^3 - 3x - 6 = 0$ by Newton-Raphson method correct to five decimal places.	CO2	6	
C)	A car covers 310 km with some speed. If its speed is increased by 10 km/hr, it will cover the same distance in one hour less. Find out the algebraic equation governing the speed of the car and find the speed of the car by Bisection method.	CO2	6	
Q. 3	Solve Any Two of the following.		12	
A)	Use Cramer's rule to solve the system of equations: $5x - 7y + z = 11$, $6x - 8y - z = 15$, $3x + 2y - 6z = 7$.	CO3	6	
B)	Apply Gauss-Elimination method to find the solution of the following system: $3x + 4y + 5z = 18$, $2x - y + 8z = 13$, $5x - 2y + 7z = 20$.	CO3	6	
C)	Write a short note on partial pivoting.	CO3	6	
Q.4	Solve Any Two of the following.		12	
A)	Evaluate the integral: $\int_{-3}^3 x^4 \cdot dx$ by using (i) Trapezoidal rule (ii) Simpson's (1/3) rd rule.	CO4	6	

B)	The table below gives the velocity V of a moving particle at a time t seconds. Find the distance covered by the particle in 12 seconds. <table><tr><td>t</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>V</td><td>4</td><td>6</td><td>16</td><td>34</td><td>60</td><td>94</td><td>136</td></tr></table>	t	0	2	4	6	8	10	12	V	4	6	16	34	60	94	136	CO4	6
t	0	2	4	6	8	10	12												
V	4	6	16	34	60	94	136												
C)	Evaluate the integral: $\int_0^6 \frac{1}{(1+x)} dx$ dividing the range into 6 equal parts by Trapezoidal rule. Also verify answer with actual integration.	CO4	6																
Q. 5	Solve Any Two of the following.		12																
A)	Fit a straight line to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>3.3</td><td>4.5</td><td>6.3</td></tr></table>	x	0	1	2	3	4	y	1	1.8	3.3	4.5	6.3	CO5	6				
x	0	1	2	3	4														
y	1	1.8	3.3	4.5	6.3														
B)	Fit a curve of the form $y=ab^x$ to the data. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>151</td><td>100</td><td>61</td><td>50</td><td>20</td><td>8</td></tr></table>	x	1	2	3	4	5	6	y	151	100	61	50	20	8	CO5	6		
x	1	2	3	4	5	6													
y	151	100	61	50	20	8													
C)	Using Newton's divide difference interpolation formula, find the values of f(2), f(8) and f(15) from given table. <table><tr><td>x</td><td>4</td><td>5</td><td>7</td><td>10</td><td>11</td><td>13</td></tr><tr><td>f(x)</td><td>48</td><td>100</td><td>294</td><td>900</td><td>1210</td><td>2028</td></tr></table>	x	4	5	7	10	11	13	f(x)	48	100	294	900	1210	2028	CO5	6		
x	4	5	7	10	11	13													
f(x)	48	100	294	900	1210	2028													
*** End ***																			